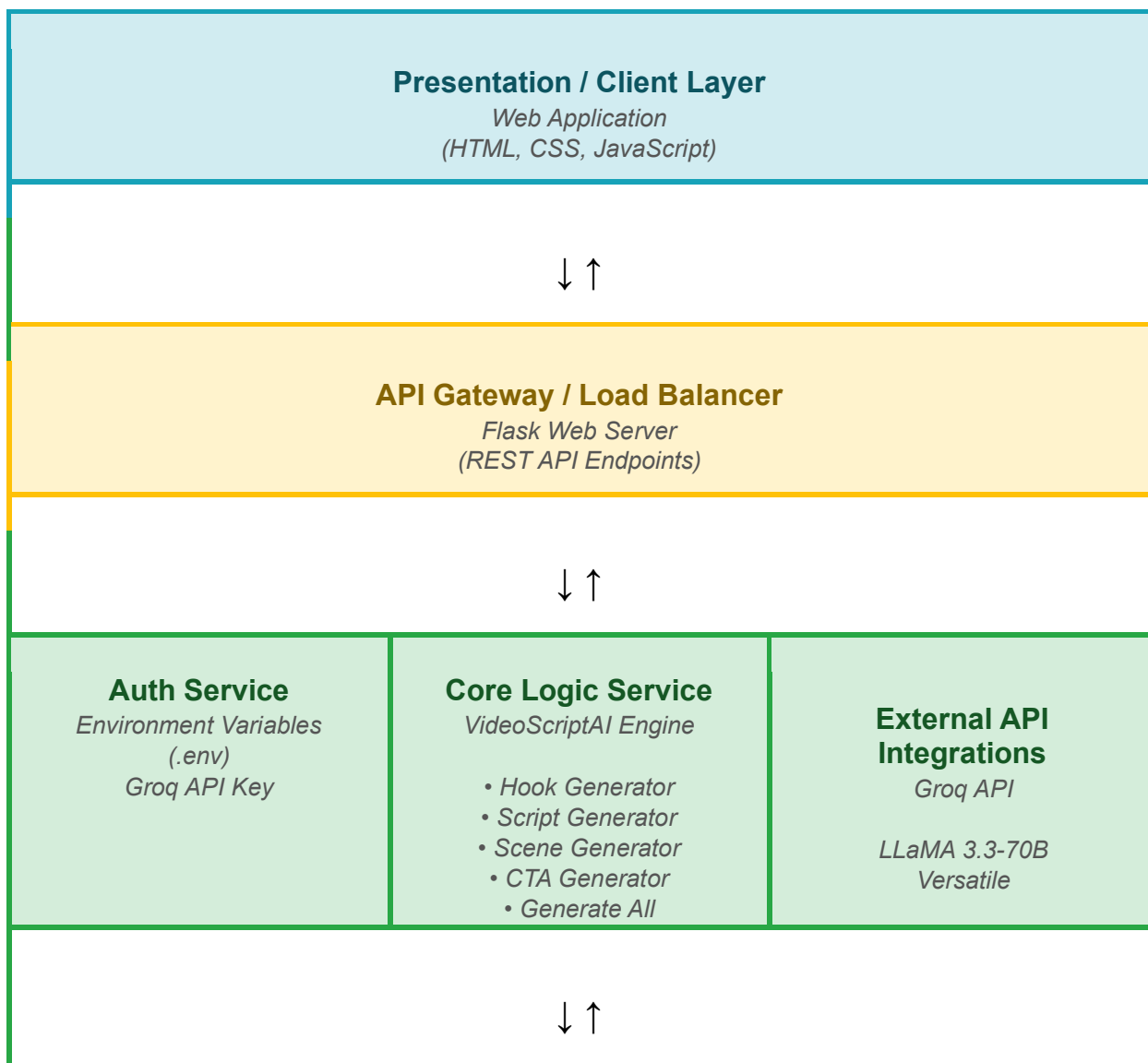


Solution Architecture

Date	15 July 2026
Team ID	SWUID2026234324
Project Name	VideoScriptAI – AI Powered Video Content Generator
Maximum Marks	5 Marks

Solution Architecture Diagram:



Data / Storage Layer

*Runtime Processing
(JSON Responses)*

(No Persistent Database)

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	<i>Collects user inputs such as product name, description, campaign goal, target audience, and duration, and displays AI-generated promotional content.</i>	<i>HTML5, CSS3, JavaScript</i>
<i>[API Gateway]</i>	<i>Receives HTTP requests from the frontend, validates input, routes requests to the appropriate AI generation functions, and returns JSON responses.</i>	<i>Python, Flask, Flask-CORS</i>
<i>[Microservices]</i>	<i>Generates Hooks, Scripts, Scene Suggestions, CTAs, and combined promotional content by constructing prompts and communicating with the Groq API.</i>	<i>Python, Prompt Engineering</i>
<i>[Database]</i>	<i>Stores configuration such as API credentials securely using environment variables. Generated content is processed in memory and returned directly to the frontend without persistent storage.</i>	<i>.env, JSON</i>